

AGREEGATE FUNCTION

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The screenshot displays the SQL Developer interface with the following components:

- SQL Commands Window:** Contains the SQL script for creating a database and table.
- Output Window:** Shows the execution results of the SQL commands.
- Snippets Panel:** Provides a list of SQL snippets for quick reference.

SQL Script:

```
-- Step 1: Create Database and Table
CREATE DATABASE aggregate_bank_db;
USE aggregate_bank_db;
CREATE TABLE bank_transactions (
    txn_id INT PRIMARY KEY,
    customer_name VARCHAR(50),
    branch_name VARCHAR(50),
    transaction_type VARCHAR(20),
    amount DECIMAL(10,2),
    transaction_date DATE
);
```

Output Window:

#	Time	Action	Message
73	21:43:56	USE aggregate_bank_db	0 row(s) affected
74	21:43:56	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, c...	0 row(s) affected

SQL Script (Continued):

```
-- Step 2: Insert Data
INSERT INTO bank_transactions VALUES
(101, 'Hima', 'Hyderabad', 'Deposit', 5000, '2024-01-05'),
(102, 'Rishika', 'Hyderabad', 'Withdrawal', 2000, '2024-01-06'),
(103, 'chowdary', 'Vijayawada', 'Deposit', 12000, '2024-01-08'),
(104, 'Narendra', 'Vizag', 'Deposit', 8000, '2024-01-10'),
(105, 'Sujatha', 'Hyderabad', 'Withdrawal', 3500, '2024-01-11'),
(106, 'Niharika', 'Vizag', 'Deposit', 15000, '2024-01-12'),
(107, 'Kutti', 'Vijayawada', 'Withdrawal', 1000, '2024-01-13'),
(108, 'Lokesh', 'Hyderabad', 'Deposit', 9000, '2024-01-14'),
(109, 'Pavani', 'Vizag', 'Withdrawal', 4000, '2024-01-15'),
(110, 'Kalayan', 'Vijayawada', 'Deposit', 11000, '2024-01-16');
```

Output Window (Continued):

#	Time	Action	Message	Duration /
74	21:43:56	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, c...	0 row(s) affected	0.015 sec
75	21:57:07	INSERT INTO bank_transactions VALUES (101, 'Hima', 'Hyderabad', 'D...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.000 sec

Snippets Panel:

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```
25 (109, 'Pavani', 'Vizag', 'Withdrawal', 4000, '2024-01-15'),
26 (110, 'Kalayan', 'Vijayawada', 'Deposit', 11000, '2024-01-16');
27 -- Step 3: SUM()
28 • SELECT SUM(amount) AS Total_Amount
29 FROM bank_transactions;
30
```

Result Grid

Total_Amount
70500.00

Result 1 x Read Only Context Help

Output

Action Output

#	Time	Action	Message
✓ 75	21:57:07	INSERT INTO bank_transactions VALUES (101, 'Hima', 'Hyderabad', 'D...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
✓ 76	21:57:46	SELECT SUM(amount) AS Total_Amount FROM bank_transactions LIM...	1 row(s) returned

```
29 FROM bank_transactions;
30 -- Step 4: AVG()
31 • SELECT AVG(amount) AS Average_Transaction
32 FROM bank_transactions;
33
```

Result Grid

Average_Transaction
7050.000000

Result 2 x Read Only Context Help

Output

Action Output

#	Time	Action	Message
✓ 76	21:57:46	SELECT SUM(amount) AS Total_Amount FROM bank_transactions LIM...	1 row(s) returned
✓ 77	21:58:26	SELECT AVG(amount) AS Average_Transaction FROM bank_transactio...	1 row(s) returned

```
33 -- Step 5: MAX()
34 • SELECT MAX(amount) AS Highest_Transaction
35 FROM bank_transactions;
36
```

Result Grid

Highest_Transaction
15000.00

Result 3 x Read Only

Output

Action Output

#	Time	Action	Message
✓ 77	21:58:26	SELECT AVG(amount) AS Average_Transaction FROM bank_transactio...	1 row(s) returned
✓ 78	21:59:20	SELECT MAX(amount) AS Highest_Transaction FROM bank_transactio...	1 row(s) returned

34

SELECT MAX(amount) AS highest_transaction

35

FROM bank_transactions;

36

-- Step 6: MIN()

37

SELECT MIN(amount) AS Lowest_Transaction

38

FROM bank_transactions;

39

40

41

disable manual current toggle

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

Lowest_Transaction
1000.00

Result Grid

Result 4

Read Only

Context Help

Output

Action Output

#	Time	Action	Message
78	21:59:20	SELECT MAX(amount) AS Highest_Transaction FROM bank_transaction...	1 row(s) returned
79	22:00:07	SELECT MIN(amount) AS Lowest_Transaction FROM bank_transaction...	1 row(s) returned

Limit to 1000 rows

43

SELECT SUM(amount) AS Total_Deposit

44

FROM bank_transactions

45

WHERE transaction_type = 'Deposit';

46

-- Step 9: GROUP BY + SUM()

47

SELECT branch_name,

48

SUM(amount) AS Total_Amount

49

FROM bank_transactions

50

GROUP BY branch_name;

51

--

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Result Grid

Filter Rows:

Export:

Wrap Cell Content:

branch_name	Total_Amount
Hyderabad	19500.00
Vijayawada	24000.00
Vizag	27000.00

Result Grid

Result 5

Result 6

Result 7

Read Only

Context Help

Output

Action Output

#	Time	Action	Message
81	22:01:18	SELECT SUM(amount) AS Total_Deposit FROM bank_transactions WH...	1 row(s) returned
82	22:01:18	SELECT branch_name, SUM(amount) AS Total_Amount FROM ban...	3 row(s) returned

Automatic context help disabled. Use the toolbar to manually get help for the current caret position or toggle automatic help.

```

59
60 -- Step 11: GROUP BY + ORDER BY
61
62 • SELECT branch_name,
63       SUM(amount) AS Total_Amount
64 FROM bank_transactions
65 GROUP BY branch_name
66 ORDER BY Total_Amount DESC;
67
--

```

Result Grid

branch_name	Total_Amount
Vizag	27000.00
Vijayawada	24000.00
Hyderabad	19500.00

Result 8

Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
82	22:01:18	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions	3 row(s) returned	0.000 sec / 0.000 sec
83	22:02:10	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions	3 row(s) returned	0.000 sec / 0.000 sec

manually get help for the current caret position or toggle automatic help.

```

78 WHERE transaction_type = 'Withdrawal';
79 -- Step 14: WHERE + AVG()
80 • SELECT AVG(amount) AS Avg_Deposit
81 FROM bank_transactions
82 WHERE transaction_type = 'Deposit';
83
--

```

Result Grid

Avg_Deposit
10000.000000

Result 9

Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
83	22:02:10	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions	3 row(s) returned	0.000 sec / 0.000 sec
84	22:02:59	SELECT AVG(amount) AS Avg_Deposit FROM bank_transactions WHERE transaction_type = 'Deposit'	1 row(s) returned	0.000 sec / 0.000 sec

Limit to 1000 rows

Jump to

74 ORDER BY Total_Transactions DESC;

75 -- Step 13: WHERE + COUNT()

76 • SELECT COUNT(*) AS Withdrawals

77 FROM bank_transactions

78 WHERE transaction_type = 'Withdrawal';

79 -- Step 14: WHERE + AVG()

80 • SELECT AVG(amount) AS Avg_Deposit

81 FROM bank_transactions

82 WHERE transaction_type = 'Deposit';

--

Automatic context help disabled. Use the toolbar manually get help for the current caret position or toggle automatic help

Result Grid

Filter Rows:

Export: Wrap Cell Content: [Context Help](#)

branch_name	Highest_Amount
Hyderabad	9000.00
Vijayawada	12000.00
Vizag	15000.00

Result 10 x Result 11 bank_transactions 12

Read Only

Context Help

Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
86	22:04:05	SELECT branch_name, MIN(amount) AS Lowest_Amount FROM ba...	3 row(s) returned	0.000 sec / 0.000 sec
87	22:04:05	SELECT * FROM bank_transactions WHERE amount > 8000 ORDER B...	4 row(s) returned	0.000 sec / 0.000 sec

Limit to 1000 rows

Jump to

89 • SELECT branch_name,

90 MIN(amount) AS Lowest_Amount

91 FROM bank_transactions

92 GROUP BY branch_name;

93 -- Step 17: WHERE + ORDER BY

94 • SELECT *

95 FROM bank_transactions

96 WHERE amount > 8000

97 ORDER BY amount DESC;

--

Auto disabled manually get help for the current caret position or toggle automatic help

Result Grid

Filter Rows:

Edit: Export/Import: Wrap Cell Content: [Context Help](#)

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date
106	Niharika	Vizag	Deposit	15000.00	2024-01-12
103	chowdary	Vijayawada	Deposit	12000.00	2024-01-08
110	Kalayan	Vijayawada	Deposit	11000.00	2024-01-16
108	Lokesh	Hyderabad	Deposit	9000.00	2024-01-14

Result 10 Result 11 bank_transactions 12 x

Apply

Context Help

Output

Action Output

#	Time	Action	Message
86	22:04:05	SELECT branch_name, MIN(amount) AS Lowest_Amount FROM ba...	3 row(s) returned
87	22:04:05	SELECT * FROM bank_transactions WHERE amount > 8000 ORDER B...	4 row(s) returned

```
103 -- Step 19: HAVING + COUNT()
104 • SELECT branch_name,
105       COUNT(*) AS Total_Transactions
106 FROM bank_transactions
107 GROUP BY branch_name
108 HAVING COUNT(*) > 3;
109
110
```

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to

branch_name	Total_Transactions
Hyderabad	4

Result 13 Result 14 x Read Only Context

Output

Action Output

#	Time	Action	Message
✓ 88	22:05:26	SELECT branch_name, COUNT(*) AS Customer_Count FROM ban...	3 row(s) returned
✓ 89	22:05:26	SELECT branch_name, COUNT(*) AS Total_Transactions FROM b...	1 row(s) returned

```
119 GROUP BY branch_name
120 HAVING SUM(amount) > 15000;
121 -- Step 22: GROUP BY + HAVING + ORDER BY
122 • SELECT branch_name,
123       AVG(amount) AS Avg_Amount
124 FROM bank_transactions
125 GROUP BY branch_name
126 HAVING AVG(amount) > 7000
127 ORDER BY Avg_Amount DESC;
...
```

branch_name	Avg_Amount
Vizag	9000.000000
Vijayawada	8000.000000

Result 15 Result 16 Result 17 x Read Only

Output

Action Output

#	Time	Action	Message
✓ 91	22:07:02	SELECT branch_name, SUM(amount) AS Total_Deposit FROM ban...	2 row(s) returned
✓ 92	22:07:02	SELECT branch_name, AVG(amount) AS Avg_Amount FROM bank...	2 row(s) returned